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Data Handling

CO2- Assignment-1

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① Mapping Data onto Aesthetics

(a) Scatterplot using mtcars dataset

```
library (ggplot2)
ggplot (mtcars, aes (wt, mpg)
  color = factor (cyl), size = hp))
    geom_point () +
    labs (title = "weight vs mileage",
  x = "Weight", y = "Mileage",
        color = "Cylinders",
      size = "Horsepower")
```

Interpretation: The Scatterplot shows the relationship between vehicle weight and mileage. Heavier vehicles generally have lower mileage.

(b) Classification of mpg dataset variables

Variable	Type	Suitable Aesthetic
manufacturer	Qualitative	color
model	Qualitative	shape
displ	Quantitative	size

year	Quantitative	X-axis
cyl	Quantitative	Size
trans	Quatitative	Colour

Interpretation: The mpg datast contains both Quantitative and qualitative variables. Different asthetics help represent these variables.

(2) Q2) Coordinate Systems and Axes

(a) Cartesian coordinates using iris

```
library (ggplot2)
ggplot (iris, aes (Sepal. length,
Petal. length)) +
  geom_point () +
  geom_abline (slope=1, intercept=0, color="red") +
  coord_fixed (ratio=1) +
  labs(title = "Sepal vs Petal length")
```

Interpretation: The graph compares sepal and petal lengths using equal scaling. The red line represents $y=x$.

(b) Polar coordinates using airquality

```
ggplot (airquality, aes (Day, Temp,
color = factor (Month)))
  geom_point() +
  coord_polar () +
  labs (title = "Daily Temps by Month")
```

-(3)

Interpretation: Polar coordinates represent data in a circular format.

(c) Cox comb chart using mpg

Interpretation : The polar bar chart displays vehicle counts for different classes in a circular form.

(3) Non linear Axes

(a) Brain weight vs body weight

```
library (ggplot2)
library (ggplot2)
ggplot (msleep, aes (bodywt, brainwt)) +
 geom_point () +
 labs (title = "Linear Scale")
ggplot (msleep, aes (bodywt, brainwt)) +
    geom_point () +
    scale_x_log10 () +
    scale_y_log10 () +
    labs (title = "log-log Scale")
```

Interpretation : Linear scales show original values while log-log scales.

(b) Boxplots of diamond prices

```
ggplot (diamonds, aes (cut, price)) +
```

Interpretation : The square root scale reduces the effect of large values and makes the distribution easier to examine. It is useful for positively skewed data.

(4) Color Scales

(a) Qualitative color palette

ggplot (iris, aes (Sepal.Length, Sepal.Width, color = Species)) + geom_point ()

Interpretation : Different colors distinguish the three iris species. The palette is designed to be colorblind friendly.

(b) Sequential Viridis heatmap

Interpretation : The heatmap uses different shades to represent density values. Darker and brighter shades.

(c) Correlation matrix heatmap

Library (ggplot2)

Interpretation : The heatmap displays correlations between vehicle variables. positive and negative values are represented.

(5) Color as a Tool to Highlight

(a) Highlight cars with mpg greater than 30

```
library (ggplot2)
mtcars$car <- rownames(mtcars)
ggplot (mtcars, aes (wt, mpg)) +
 geom_point ( color = "grey") +
 geom_point (data = subset (mtcars, mpg > 30)
```

Interpretation : Cars with mileage greater than 30 are highlighted in red. Other cars are shown in grey.

(b) Continuous and discrete color scales

Interpretation :

- Continuous scale represents numerical values using a color gradient
- Discrete scale represents seperate categories using different colors.

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Output

